DP Approach for SPOJ Mixtures (Handwritten Notes)

Dynamic Programming: SPOJ Mixtures Problem

Problem Statement (Summary):

Given n mixtures with initial colors.

Goal: find min total smoke by mixing all to one.

Define $dp(i, j)$ as minimum smoke to mix mixtures from index i to j.

$$\text{smoke} = \left(\sum_{i}^{k}\right) \cdot \left(\sum_{k+1}^{j}\right)$$

A(B(CD))

i k k+1 j

New Color: $(\text{color}_a + \text{color}_b) \bmod 100$

Optimal substructure

$$dp(i, j) = \min_{i \le k < j} \left\{ dp(i, k) + dp(k+1, j) + \left(\text{Color}_{sum}(i,k) * \text{Color}_{sum}(k+1,j)\right) \right\}$$

Complexity

Time: $O(n^3)$

Space: $O(n^2)$